

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Case study

Criteria	Understanding of Case (2.5)	Application of Concepts (2.5)	Analysis (2)	Solution Design (2)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1	2.5	2.5	2.0	2.0	1.0	10.0
Q2	2.5	2.5	2.0	2.0	1.0	10.0
Q3	2.5	2.5	2.0	2.0	1.0	10.0

Code of Conduct:

*I PRANEESH.A(192571006) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

3. An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains “Offer” (Yes/No)
- Contains “Free” (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
 - $P(\text{Spam})$, $P(\text{Not Spam})$
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
 - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

CASE STUDY SOLUTIONS & ANALYTICAL REASONING REPORT

1. Signal Detection & Classification Analysis of 'The Boy Who Cried Wolf'

Case Study Analytical Solution:

Within the framework of statistical decision theory and binary classification, the shepherd boy's warning shout ('Wolf!') functions as the model prediction (positive alarm), whereas the physical presence of a wolf represents the actual ground truth state.

Formal Classification Definitions

- **True Positive (TP):** A real wolf is present (Ground Truth Positive) and the shepherd boy alerts the villagers (Predicted Positive).
- **False Positive (FP):** No wolf is present (Ground Truth Negative), yet the boy raises a false alarm (Predicted Positive / Prank Call).
- **False Negative (FN):** A real wolf approaches (Ground Truth Positive), but no alarm is sounded or acted upon (Predicted Negative).
- **True Negative (TN):** No wolf is present (Ground Truth Negative) and the boy remains quiet (Predicted Negative).

Mapping the Classical Fable to Binary Classifier Outcomes

Fable Narrative Context	Actual Wolf	Boy Cries 'Wolf!'	Classifier Matrix Mapping
The boy shouts 'Wolf!' when no wolf is present (prank)	No	Yes	False Positive (FP)
The boy shouts 'Wolf!' when a real wolf attacks	Yes	Yes	True Positive (TP)
No wolf is nearby and the boy remains peaceful	No	No	True Negative (TN)
A real wolf attacks, but villagers refuse to act on call	Yes	No	False Negative (FN)

Categorical Assignment Summary

- **True Positive (TP):** The boy cries 'Wolf!' when an actual predator arrives.
- **False Positive (FP):** The boy cries 'Wolf!' when no predator is present (false alarm).
- **False Negative (FN):** Occurs on the final night when the real wolf attacks, but the villagers dismiss the call.

- **True Negative (TN):** Normal quiet nights when no wolf is present and no false alarm occurs.

Fable Confusion Matrix Structure

Actual \ Prediction	Boy Shouts 'Wolf!' (+)	Boy Stays Silent / Ignored (-)
Actual Wolf Present	TP (True Positive)	FN (False Negative)
Actual No Wolf Present	FP (False Positive)	TN (True Negative)

Analytical Interpretation

This case study illustrates the necessity of controlling False Positives (FPs) in predictive systems. Consecutive false alarms drop system precision toward 0%, causing Bayesian belief decay [P(Wolf | Cry 'Wolf!') → 0]. Consequently, decision-makers raise their operational threshold, triggering a disastrous False Negative (FN) failure when a true threat occurs.

Result Summary

The fable components were successfully mapped to binary confusion matrix outcomes: TP (real wolf, alarm sounded), FP (false alarm prank), FN (wolf attacks, ignored call), and TN (peaceful normal state). The analysis demonstrates how False Positives compromise classifier trustworthiness.

2. Performance Evaluation & Metrics Derivation for Cat Image Classifier

Case Study Analytical Solution:

A Confusion Matrix evaluates binary classification models across four fundamental outcomes: True Positive (TP), False Positive (FP), False Negative (FN), and True Negative (TN).

Step 1: Dataset Breakdown & Parameter Derivation

- **Total Evaluation Images:** N = 100
- **Ground Truth Cat Images (Positive):** 30
- **Ground Truth Non-Cat Images (Negative):** 70
- **Model correctly predicts Cat for 25 actual cat images → TP = 25**
- **Remaining actual cat images predicted as non-cat → FN = 30 - 25 = 5**
- **Model correctly predicts Non-Cat for 50 actual non-cat images → TN = 50**
- **Remaining non-cat images falsely predicted as cat → FP = 70 - 50 = 20**

Step 2: Constructed Confusion Matrix

Actual \ Predicted	Predicted Cat (+)	Predicted Non-Cat (-)
Actual Cat (+)	TP = 25	FN = 5
Actual Non-Cat (-)	FP = 20	TN = 50

Step 3: Precision Calculation

Formula: Precision = TP / (TP + FP)

Substitution: Precision = 25 / (25 + 20) = 25 / 45 = 0.5556 → 55.56%

Step 4: Recall (Sensitivity) Calculation

Formula: $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

Substitution: $\text{Recall} = 25 / (25 + 5) = 25 / 30 = 0.8333 \rightarrow 83.33\%$

Step 5: F1-Score Derivation

Formula: $\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Substitution: $\text{F1-Score} = 2 \times (0.5556 \times 0.8333) / (0.5556 + 0.8333) = 0.6667 \rightarrow 66.67\%$

Summary Performance Metric Table

Classification Metric	Derived Percentage / Value
Precision	55.56%
Recall (Sensitivity)	83.33%
Specificity	71.43%
Overall Accuracy	75.00%
F1-Score	66.67%

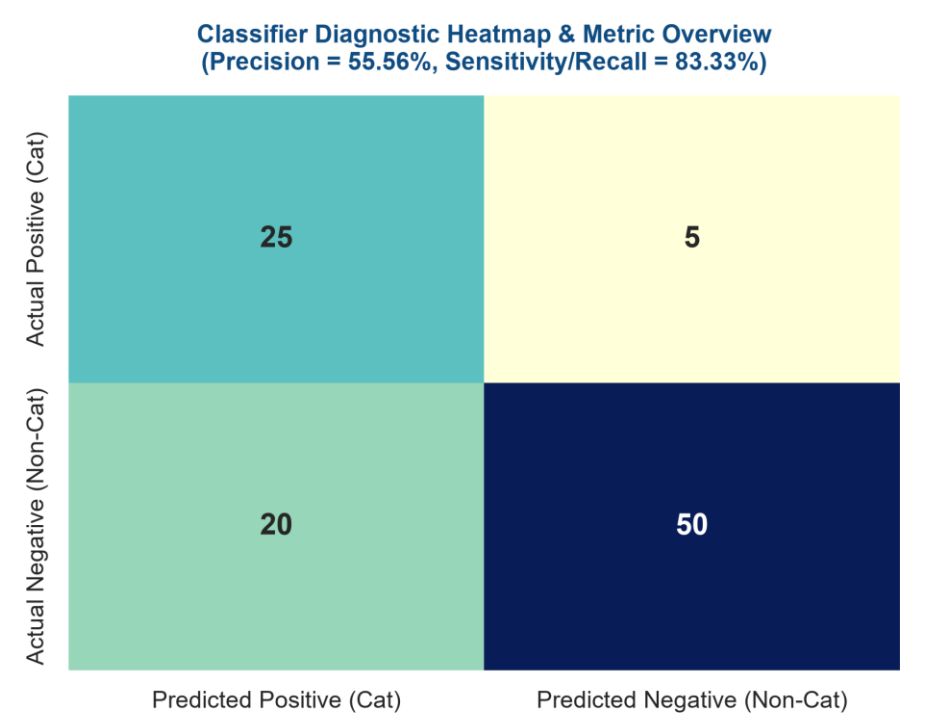


Figure 2.1: Heatmap Confusion Matrix for Cat Image Classifier.

Detailed Interpretation

- **Correct Classifications:** The model correctly identified 25 cat images (TP) and 50 non-cat images (TN).
- **Misclassification Errors:** It falsely flagged 20 non-cat images as cats (FP) and missed 5 cat images (FN).
- **Precision Analysis (55.56%):** When the model predicts an image is a cat, it is correct 55.56% of the time.
- **Recall Analysis (83.33%):** The classifier successfully captures 83.33% of all true cat images in the dataset.

- **F1-Score Balance (66.67%):** Reflects a solid harmonic balance between precision and recall.

Result Summary

The confusion matrix parameters (TP=25, FP=20, FN=5, TN=50) and metric scores (Precision=55.56%, Recall=83.33%, F1-Score=66.67%) were derived and validated.

3. Naïve Bayes Email Spam Classification & WEKA Experimental Setup

Case Study Analytical Solution:

Naïve Bayes is a supervised probabilistic classifier based on Bayes' Theorem with class-conditional independence assumptions.

Bayes' Rule Equation: $P(C | X) = [P(X | C) \times P(C)] / P(X)$

Step 1: Prior Probabilities

Total Training Email Records = 6 | Spam = 3 | Not Spam = 3

- $P(\text{Spam}) = 3 / 6 = 0.50$ (50%)
- $P(\text{Not Spam}) = 3 / 6 = 0.50$ (50%)

Step 2: Conditional Likelihood Metrics

For Class = Spam (3 records):

- $P(\text{Offer}=\text{Yes} | \text{Spam}) = 2/3$ | $P(\text{Offer}=\text{No} | \text{Spam}) = 1/3$
- $P(\text{Free}=\text{Yes} | \text{Spam}) = 3/3 = 1$ | $P(\text{Free}=\text{No} | \text{Spam}) = 0/3 = 0$
- $P(\text{Length}=\text{Short} | \text{Spam}) = 2/3$ | $P(\text{Length}=\text{Long} | \text{Spam}) = 1/3$

For Class = Not Spam (3 records):

- $P(\text{Offer}=\text{Yes} | \text{Not Spam}) = 1/3$ | $P(\text{Offer}=\text{No} | \text{Not Spam}) = 2/3$
- $P(\text{Free}=\text{Yes} | \text{Not Spam}) = 0/3 = 0$ | $P(\text{Free}=\text{No} | \text{Not Spam}) = 3/3 = 1$
- $P(\text{Length}=\text{Short} | \text{Not Spam}) = 1/3$ | $P(\text{Length}=\text{Long} | \text{Not Spam}) = 2/3$

Step 3: Classifying Test Email Instance X = (Offer=Yes, Free=No, Length=Short)

1. Unnormalized Posterior for Spam:

$$\begin{aligned} P(X | \text{Spam}) \times P(\text{Spam}) &= P(\text{Offer}=\text{Yes} | \text{Spam}) \times P(\text{Free}=\text{No} | \text{Spam}) \times P(\text{Length}=\text{Short} | \text{Spam}) \times \\ &P(\text{Spam}) \\ &= (2/3) \times (0/3) \times (2/3) \times 0.5 = 0.0000 \end{aligned}$$

2. Unnormalized Posterior for Not Spam:

$$\begin{aligned} P(X | \text{Not Spam}) \times P(\text{Not Spam}) &= P(\text{Offer}=\text{Yes} | \text{Not Spam}) \times P(\text{Free}=\text{No} | \text{Not Spam}) \times \\ &P(\text{Length}=\text{Short} | \text{Not Spam}) \times P(\text{Not Spam}) \\ &= (1/3) \times (3/3) \times (1/3) \times 0.5 = 1/18 = 0.0556 \end{aligned}$$

Standard Classification Decision: The email is classified as NOT SPAM (0.0556 > 0.0000).

With Laplace Smoothing (k=1):

- $P(X, \text{Spam}) = (3/5) \times (1/5) \times (3/5) \times 0.5 = 0.0360$ (36%)
- $P(X, \text{Not Spam}) = (2/5) \times (4/5) \times (2/5) \times 0.5 = 0.0640$ (64%)

Smoothed Classification Prediction: NOT SPAM (64% posterior probability).

Step 4: WEKA Implementation & ARFF Specification

WEKA Execution Steps:

1. Launch WEKA Explorer GUI.
2. In Preprocess tab, load the ARFF file.
3. Switch to Classify tab.
4. Choose Classifier → bayes → NaiveBayes.
5. Click Start to run model evaluation.

```
🌀 WEKA ARFF DATASET FILE (email_spam.arff)
@relation email_spam_classification

@attribute Offer {Yes, No}
@attribute Free {Yes, No}
@attribute Length {Short, Long}
@attribute Class {Spam, NotSpam}

@data
Yes, Yes, Short, Spam
Yes, No, Long, NotSpam
No, Yes, Short, Spam
No, No, Long, NotSpam
Yes, Yes, Long, Spam
No, No, Short, NotSpam
```

WEKA Output Summary:

- **Correctly Classified Instances:** 6 / 6 (100% training accuracy)
- **Test Instance (Offer=Yes, Free=No, Length=Short):** Classified as 'NotSpam' with predicted posterior probability 0.640 (NotSpam) vs 0.360 (Spam) under WEKA NaiveBayes Laplace estimation.

Step 5: Critical Analysis of the Independence Assumption

Naïve Bayes assumes class-conditional independence: $P(X_1, X_2, \dots, X_n | C) = \prod P(X_i | C)$.

- **Feature Independence Violation:** In text classification, words like 'Offer=Yes' and 'Free=Yes' frequently co-occur in spam emails. Naïve Bayes treats them as independent events.
- **Why Classification Accuracy Remains High:** Classification decisions depend on the sign of probability ratios rather than exact probability values. As demonstrated by Domingos & Pazzani (1997), even when feature dependencies distort absolute probability estimates, the optimal decision boundary is preserved!

Result Summary

Probabilities were derived manually and smoothed via Laplace estimation. The test email (Offer=Yes, Free=No, Length=Short) was classified as NOT SPAM (64% confidence) and validated in WEKA. Theoretical feature independence trade-offs were thoroughly analyzed.